

A* vs Dijkstra - Shortest Path

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In the example from the book, Dijkstra's Algorithm finds the shortest path from Arad to Bucharest by always choosing the node with the lowest cost from the starting point. Its evaluation function is $f(n) = g(n)$. The optimal path is Arad -> Sibiu -> Rimnicu Vilcea -> Pitesti -> Bucharest, with a total cost of 418 km.

A* uses both the cost already traveled and a heuristic that estimates the remaining distance to Bucharest. Its evaluation function is $f(n) = g(n) + h(n)$. Because the heuristic gives information about which cities are closer to the destination, A* avoids exploring some unnecessary nodes and reaches the same optimal path with a cost of 418 km.

Therefore, A* is better for this example because it finds the same optimal solution while exploring fewer nodes. Dijkstra can be seen as a special case of A* where $h(n) = 0$, meaning that it has no information about the destination.